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# Does AI Foster imposter feelings? The impact of task design on students' use of AI

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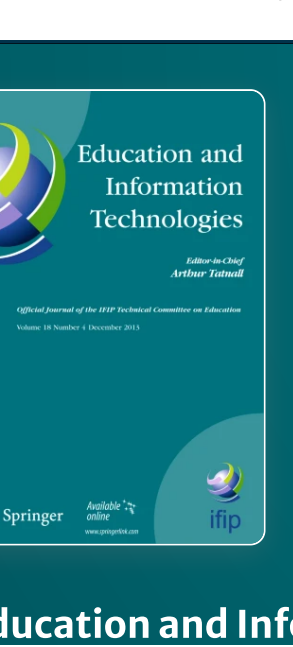

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## Abstract

The adoption of artificial intelligence (AI) in higher education is transforming the way students experience the learning process. The ability of AI to complete academic tasks in seconds and with high quality makes it an appealing tool for students. However, teachers have not adapted their tasks to the use of this tool. Inappropriate use of AI by students—using it as a replacement for their own learning process—can negatively impact their learning experience. This study examines how task design—whether delegable or supported—affects students' learning experiences when using AI. Following an experimental methodology, three studies were conducted to evaluate perceived learning, perceived effort, student agency, psychological ownership of tasks, and impostor syndrome among social sciences students. The study also explores how these perceptions evolve over time. The findings show that tasks that encourage delegating resolution to AI deteriorate the learning experience and sense of task ownership by students. Moreover, the way students use AI plays a key role in the development of impostor syndrome, a phenomenon that intensifies over time. The study concludes by emphasizing that effective AI integration requires the joint commitment of both teachers and students, with specific training in the use of AI tools being essential.

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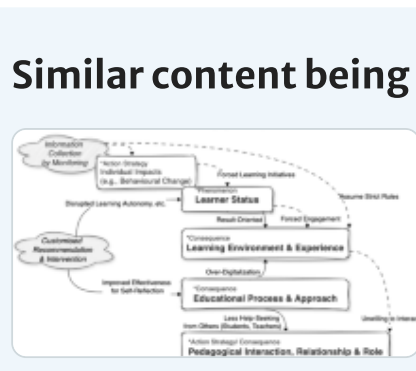

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## Data Availability

The data that support the findings of this study are available from the authors upon reasonable request.

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### Competing interests

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## Appendix

The war for the future of furniture is being fought in Madrid: https://www.elconfidencial.com/empresas/2024-09-25/ikea-segunda-mano-madrid_3968755/

IKEA holds 9% of the global furniture market, but it is losing ground in the fast-growing pre-owned sector. Since August, the company has been running a pilot program in the Spanish capital, with potential global implications.

| Delegable Task – Questions | Main cognitive operation required |
|---|---|
| 1. What strategies has IKEA implemented to differentiate itself in the secondhand furniture market? | Information retrieval and summarization |
| 2. Compare the business model of IKEA Preowned with that of platforms such as Wallapop and Vinted. What are the differences and similarities between them? | Comparison based on structured features |
| 3. How has the growing concern for sustainability influenced the perception of the IKEA brand among younger consumers? | Descriptive trend analysis |
| 4. Define the consumer profile of re-commerce platforms (listing features such as sociodemographic and psychographic characteristics). | Profiling from demographic data |
| 5. Compare pricing strategies in the primary market (IKEA) versus the secondhand market (Preowned, Vinted, or Wallapop). | Comparative numerical reasoning |
| Supported Task – Questions | |
| 1. In what ways can IKEA compete effectively in a market dominated by platforms such as Wallapop and Vinted without losing its brand identity? | Strategic evaluation and judgment |
| 2. To what extent has the "Greta effect" influenced IKEA's sales and positioning in the global market? | Interpretive analysis of public sentiment |
| 3. What impact does the shift toward re-commerce have on traditional consumer culture, and how can IKEA leverage this trend without compromising its relationship with its most loyal customers? | Cultural reasoning and trend adaptation |
| 4. How would you evaluate IKEA's market testing strategy with its Preowned platform? Consider the types of data collected, the duration of the pilot, and the overall test design in your analysis. | Critical evaluation of experimental design |
| 5. Based on the price data of the four products sold on re-commerce platforms, explain why purchasing Wallapop items is, in some cases, more expensive than purchasing new items directly from IKEA. | Understanding consumer anomalies |

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